



AI Attention Visualizer

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- **Education:** Maintain student data, attendance, and grades.

Components of a Database

Databases consist of several critical components that work together to **store, organize, and retrieve data** effectively. Here's a detailed explanation of each component:

1. Data

Data is the core component of any database, representing the **actual information stored**. It can include **numbers, text, images, videos, or documents**, depending on the database's purpose. For instance, a customer database might store customer names, addresses, and purchase histories.

2. Schema

The schema is the blueprint or structure of the database. It defines how data is organized and includes details like **tables, columns, data types**, and relationships between entities. For example, a table in a customer database might have columns like **CustomerID, Name, and Email**. The schema ensures consistency and helps users understand how the database is designed.

3. DBMS

The DBMS is the **software layer** that enables interaction with the database. It manages the **storage, retrieval, and manipulation** of data while ensuring security and data integrity. Examples of DBMS software include MySQL, Oracle, and MongoDB. The DBMS also handles tasks like backup, recovery, and query optimization to maintain the database's performance.



Extracted Text

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Queries are commands used to interact with the database, allowing users to retrieve, manipulate, or



Word Attention

Education



Maintain



student



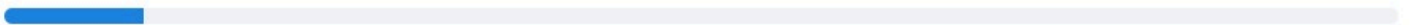
data



attendance



and



grades



Components

